

Example 3: State which rule of inference is the basis of the following argument: "It is below freezing now". Therefore it is below freezing or raining now."

Soln: p : It is below freezing now. q : It is raining now.
The argument is of the form: $\frac{p}{\therefore p \vee q}$ } Addition Rule

Example 4: State which rule of inference is the basis of the following argument: "It is below freezing and raining now. Therefore, it is below freezing now."

Soln: p : It is below freezing now q : It is raining now
The argument is of the form: $\frac{p \wedge q}{\therefore p}$ } Simplification Rule

Example 5: State which rule of inference is used in the argument:
If it rains today, then we will not have a barbecue today. If we do not have a barbecue today, then we will have a barbecue tomorrow. Therefore, if it rains today, then we will have a barbecue tomorrow.

Soln: p : It rains today q : We will not have a barbecue today
 r : We will have a barbecue tomorrow.
The argument is of the form: $\frac{p \rightarrow \neg q \quad q \rightarrow r}{\therefore p \rightarrow r}$ } Hence, this argument is a hypothetical syllogism

1.6.4 Using Rules of Inference to Build Arguments

When there are many premises, several rules of inference are often needed to show that an argument is valid.